RULE-BASED CHATBOT

AI Internship Project Report

1. Introduction

A Rule-Based Chatbot is a simple chatbot that communicates with users using predefined rules and responses. It uses Python conditional statements to identify user messages and generate suitable replies.

2. Problem Statement

The objective is to develop a basic chatbot that can understand simple user inputs and respond automatically using predefined conditions.

3. Objectives

- Understand basic chatbot development.
- Implement if-else statements in Python.
- Learn basic pattern matching.
- Create a simple conversation flow.
- Understand fundamental Natural Language Processing concepts.

4. Technologies Used

- Programming Language: Python
- Platform: Google Colab
- Concepts: If-Else, While Loop, String Matching

5. Methodology

The chatbot receives input from the user and converts it into lowercase. It checks the input against predefined conditions and generates an appropriate response.

A while loop maintains the conversation until the user enters "bye".

6. Working of the Chatbot

The chatbot responds to different messages:

- Hello: Displays a greeting.
- How are you: Gives a status response.
- Your name: Introduces itself.
- Help: Provides assistance.

- Thank you: Responds politely.
- Bye: Ends the conversation.

7. Output

The chatbot was successfully executed in Google Colab. It responded correctly to the tested user inputs and terminated when the user entered "Bye".

Output Screenshot: Insert the screenshot of the successful chatbot execution here.

8. Advantages

- Easy to develop.
- Simple to understand.
- Provides instant predefined responses.
- Requires minimal computing resources.

9. Limitations

- Cannot understand complex sentences.
- Gives only predefined responses.
- Cannot learn automatically from conversations.

10. Conclusion

The Rule-Based Chatbot project was successfully developed using Python. This project helped demonstrate basic chatbot development, conditional statements, pattern matching, and conversation flow.